Bitwise Operators

Problem Submissions Leaderboard Editorial

In this challenge, you will use logical bitwise operators. All data is stored in its binary representation. The logical operators, and C language, use $\mathbf{1}$ to represent true and $\mathbf{0}$ to represent false. The logical operators compare bits in two numbers and return true or false, $\mathbf{0}$ or $\mathbf{1}$, for each bit compared.

- `Bitwise AND operator` & The output of bitwise AND is 1 if the corresponding bits of two operands is 1. If either bit of an operand is 0, the result of corresponding bit is evaluated to 0. It is denoted by &.
- `Bitwise OR operator` | The output of bitwise OR is 1 if at least one corresponding bit of two operands is 1. It is denoted by |.
- `Bitwise XOR (exclusive OR) operator` ^ The result of bitwise XOR operator is 1 if the corresponding bits of two operands are opposite. It is denoted by $\oplus$.

For example, for integers 3 and 5,

```
3 = 00000011 (In Binary)
5 = 00000101 (In Binary)

AND operation          OR operation          XOR operation
   00000011               00000011              00000011
& 00000101             | 00000101            ^ 00000101
   --------               --------              --------
   00000001  = 1          00000111  = 7         00000110  = 6
```

You will be given an integer n, and a threshold, k. $Foreachnumber$i$from$1$through$n$, findthemaximumvalueofthelogicaland, orandxorwhencomparedagainstallintegersthrough$n$thataregreaterthan$i$. Consideravalueonlyifthecomparisonreturnsaresultlessthan$k$. Print the results of the and, or and exclusive or comparisons on separate lines, in that order.

Example

$n = 3$

$k = 3$

The results of the comparisons are below:

```
a b   and or xor
1 2   0   3  3
1 3   1   3  2
2 3   2   3  1
```

For the `and` comparison, the maximum is $\mathbf{2}$. For the `or` comparison, none of the values is less than k, so the maximum is $\mathbf{0}$. For the `xor` comparison, the maximum value less than k is $\mathbf{2}$. The function should print:

```
2
0
2
```

Function Description

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Print the maximum values for the `and`, `or` and `xor` comparisons, each on a separate line.

Input Format

The only line contains 2 space-separated integers, n and k.

Constraints

- $2 \leq n \leq 10^3$
- $2 \leq k \leq n$

Sample Input 0

```
5 4
```

Sample Output 0

```
2
3
3
```

Explanation 0

$n = 5, k = 4$

$S = \{1, 2, 3, 4, 5\}$

All possible values of a and b are:

1. $\mathrm{a} = 1, \mathrm{b} = 2;\ \mathrm{a} \,\&\, \mathrm{b} = 0;\ \mathrm{a} \mid \mathrm{b} = 3;\ \mathrm{a} \oplus \mathrm{b} = 3;$
2. $\mathrm{a} = 1, \mathrm{b} = 3;\ \mathrm{a} \,\&\, \mathrm{b} = 1;\ \mathrm{a} \mid \mathrm{b} = 3;\ \mathrm{a} \oplus \mathrm{b} = 2;$
3. $\mathrm{a} = 1, \mathrm{b} = 4;\ \mathrm{a} \,\&\, \mathrm{b} = 0;\ \mathrm{a} \mid \mathrm{b} = 5;\ \mathrm{a} \oplus \mathrm{b} = 5;$
4. $\mathrm{a} = 1, \mathrm{b} = 5;\ \mathrm{a} \,\&\, \mathrm{b} = 1;\ \mathrm{a} \mid \mathrm{b} = 5;\ \mathrm{a} \oplus \mathrm{b} = 4;$
5. $\mathrm{a} = 2, \mathrm{b} = 3;\ \mathrm{a} \,\&\, \mathrm{b} = 2;\ \mathrm{a} \mid \mathrm{b} = 3;\ \mathrm{a} \oplus \mathrm{b} = 1;$
6. $\mathrm{a} = 2, \mathrm{b} = 4;\ \mathrm{a} \,\&\, \mathrm{b} = 0;\ \mathrm{a} \mid \mathrm{b} = 6;\ \mathrm{a} \oplus \mathrm{b} = 6;$
7. $\mathrm{a} = 2, \mathrm{b} = 5;\ \mathrm{a} \,\&\, \mathrm{b} = 0;\ \mathrm{a} \mid \mathrm{b} = 7;\ \mathrm{a} \oplus \mathrm{b} = 7;$
8. $\mathrm{a} = 3, \mathrm{b} = 4;\ \mathrm{a} \,\&\, \mathrm{b} = 0;\ \mathrm{a} \mid \mathrm{b} = 7;\ \mathrm{a} \oplus \mathrm{b} = 7;$
9. $\mathrm{a} = 3, \mathrm{b} = 5;\ \mathrm{a} \,\&\, \mathrm{b} = 1;\ \mathrm{a} \mid \mathrm{b} = 7;\ \mathrm{a} \oplus \mathrm{b} = 6;$
10. $\mathrm{a} = 4, \mathrm{b} = 5;\ \mathrm{a} \,\&\, \mathrm{b} = 4;\ \mathrm{a} \mid \mathrm{b} = 5;\ \mathrm{a} \oplus \mathrm{b} = 1;$

- The maximum possible value of $a\&b$ that is also $< (k = 4)$ is 2, so we print 2 on first line.
- The maximum possible value of $a|b$ that is also $< (k = 4)$ is 3, so we print 3 on second line.
- The maximum possible value of $a \oplus b$ that is also $< (k = 4)$ is 3, so we print 3 on third line.

Change Theme Language: C

```c
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
```

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```
    int n, k;

    scanf("%d %d", &n, &k);
    calculate_the_maximum(n, k);

    return 0;
}
```

Line: 20 Col: 1

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